7-ordering: 序关系 (Ordering)

魏恒峰

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Definition (Equivalence Relation)

$R \subseteq X \times X$ is an *equivalence relation* on X iff R is

- reflexive (自反): $\forall a \in X \; aRa$
- symmetric (对称): $\forall a, b \in X \; (aRb \leftrightarrow bRa)$
- transitive (传递): $\forall a, b, c \in X \; (aRb \wedge bRc \rightarrow aRc)$

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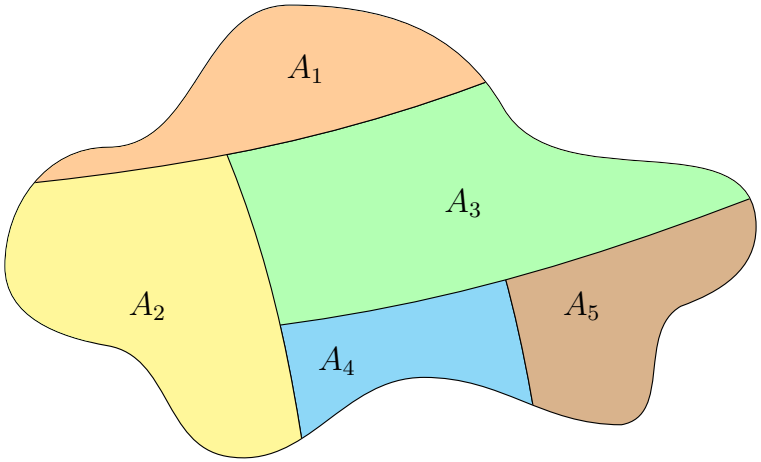

Equivalence Relation $\Longleftrightarrow$ Partition $\Longleftrightarrow$ Abstraction

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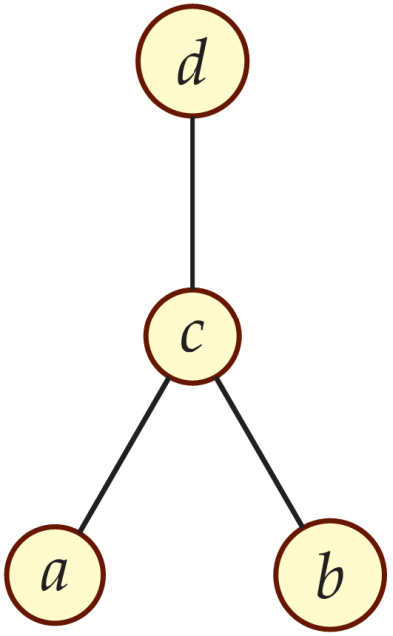
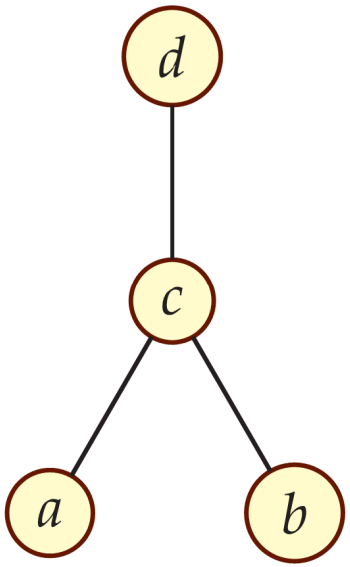


Ordering?

"*Order theory* is a branch of mathematics that investigates the *intuitive notion of order* using *binary relations*.



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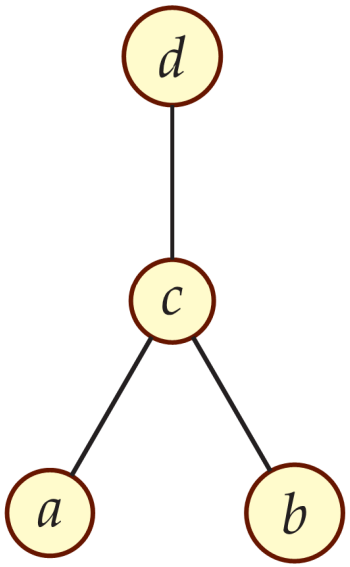

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— *Order theory (Wikipedia)*

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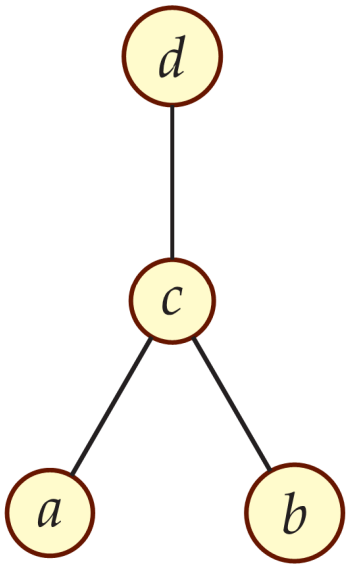

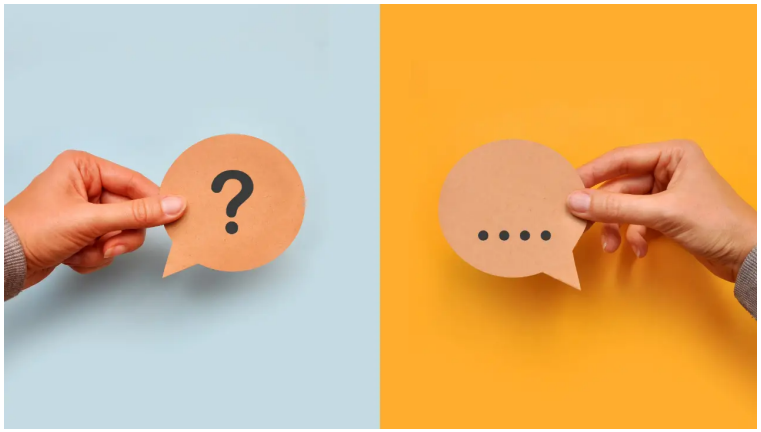

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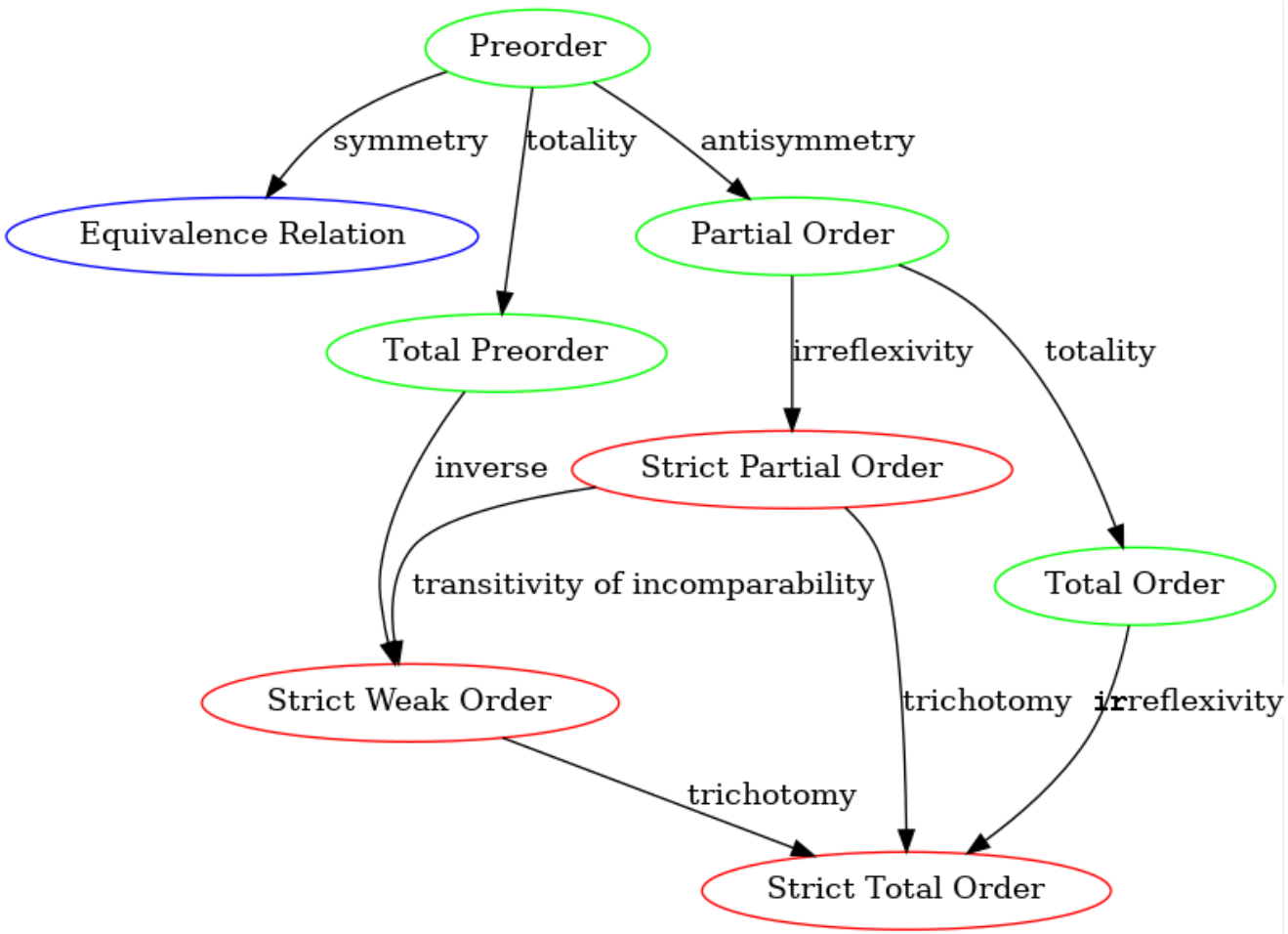
Preorder
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Definition (偏序关系 (Partial Order; Poset))

令 $\preceq \subseteq X \times X$ 是 X 上的二元关系。

如果 $\preceq$ 满足以下条件, 则称 $\preceq$ 是 X 上的偏序关系,

并称 $(X, \preceq)$ 为偏序集 (poset; Partially Ordered Set):

(1) $\preceq$ 是自反 (reflexive)的。

$$\forall x \in X \; x \preceq x.$$

(2) $\preceq$ 是**反对称 (antisymmetric)**的。

$$\forall x, y \in X \; x \preceq y \wedge y \preceq x \to x = y.$$

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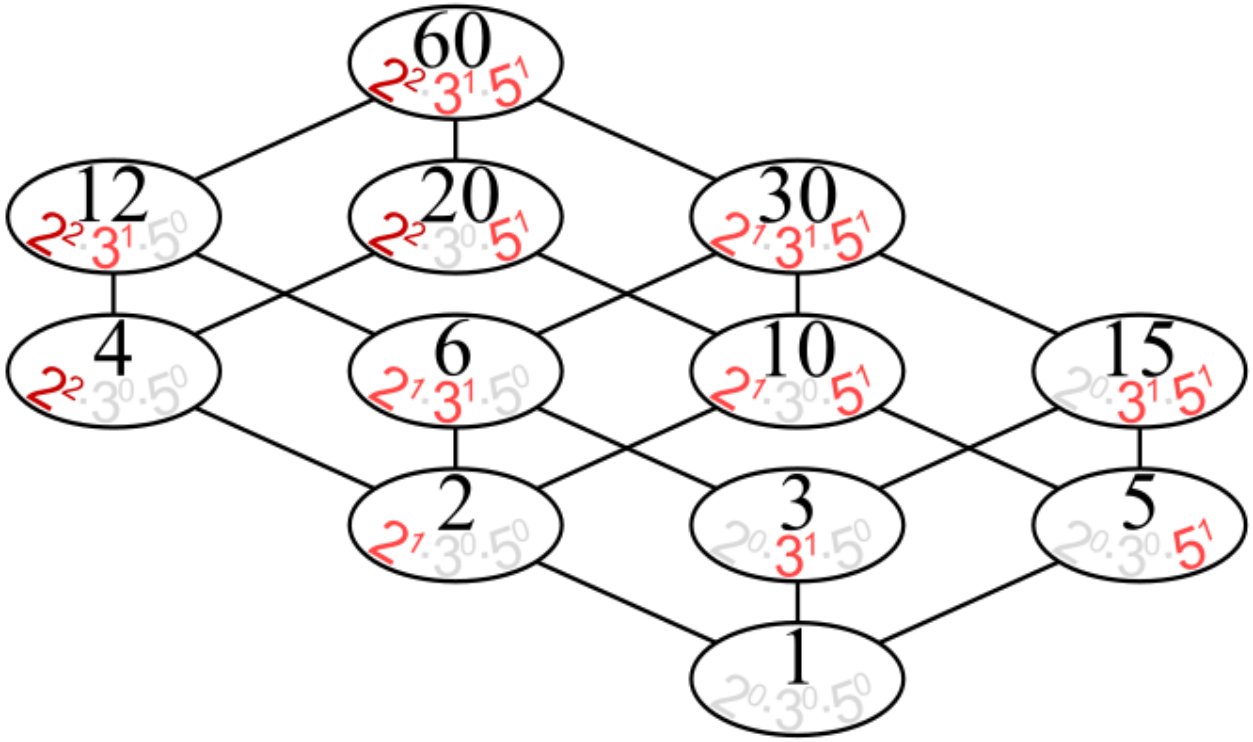
$$x \prec y \triangleq x \preceq y \wedge x \neq y$$

$$X = \{1, 2, 3, 4, 5, 6, 10, 12, 15, 20, 30, 60\}$$

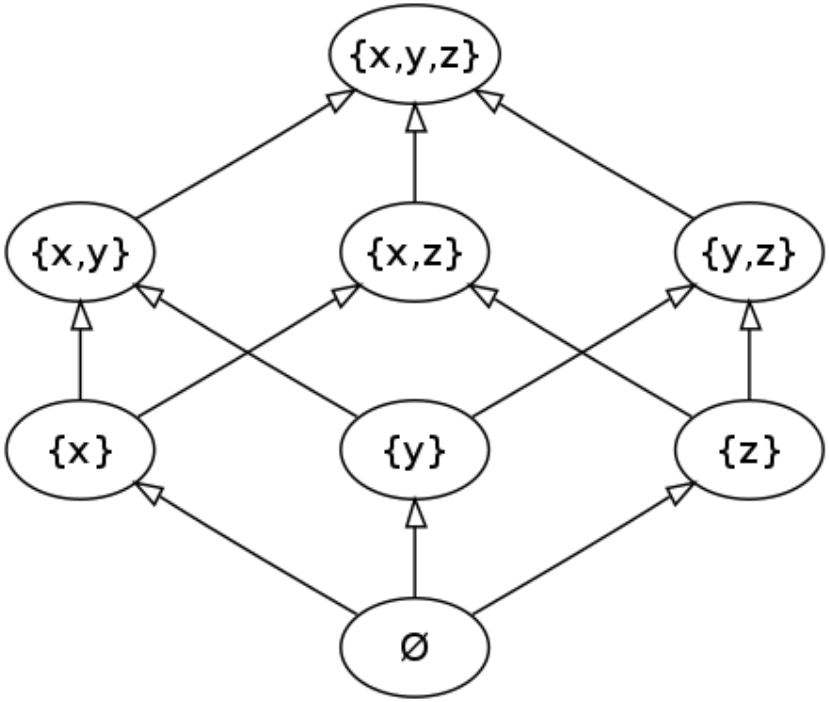
X 上的整除关系



自反 + 反对称 + 传递

$$S = \{x, y, z\}$$

$\mathcal{P}(S)$ 上的包含 $\subseteq$关系



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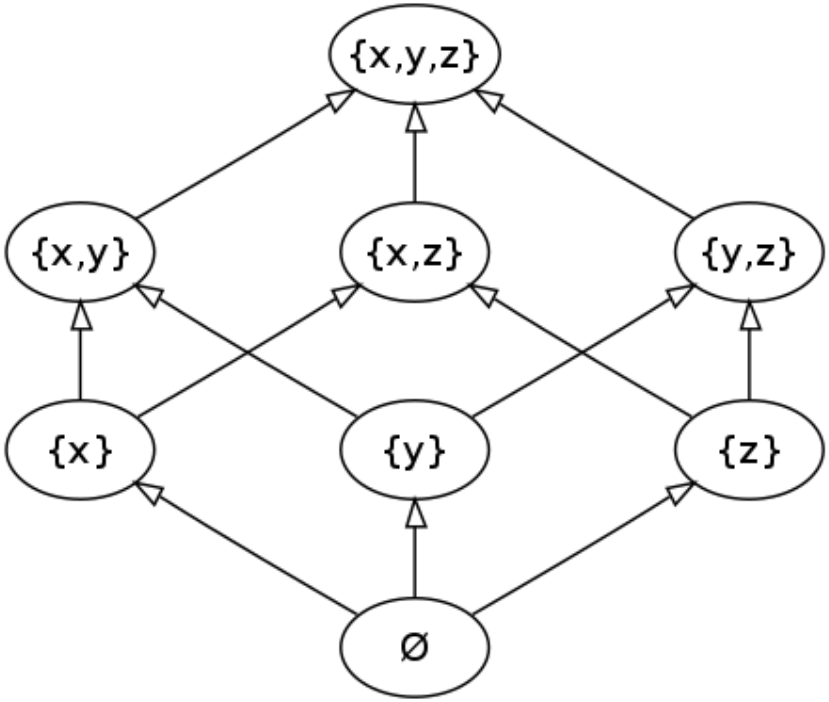
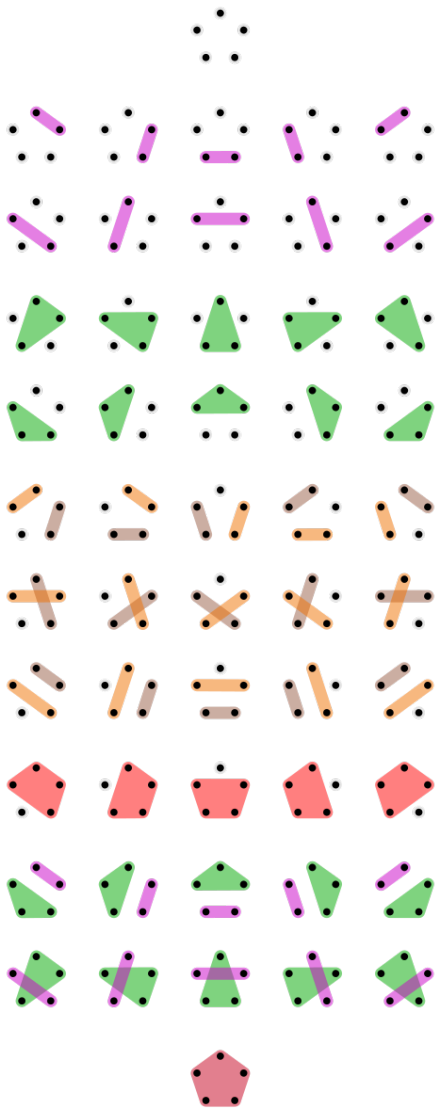
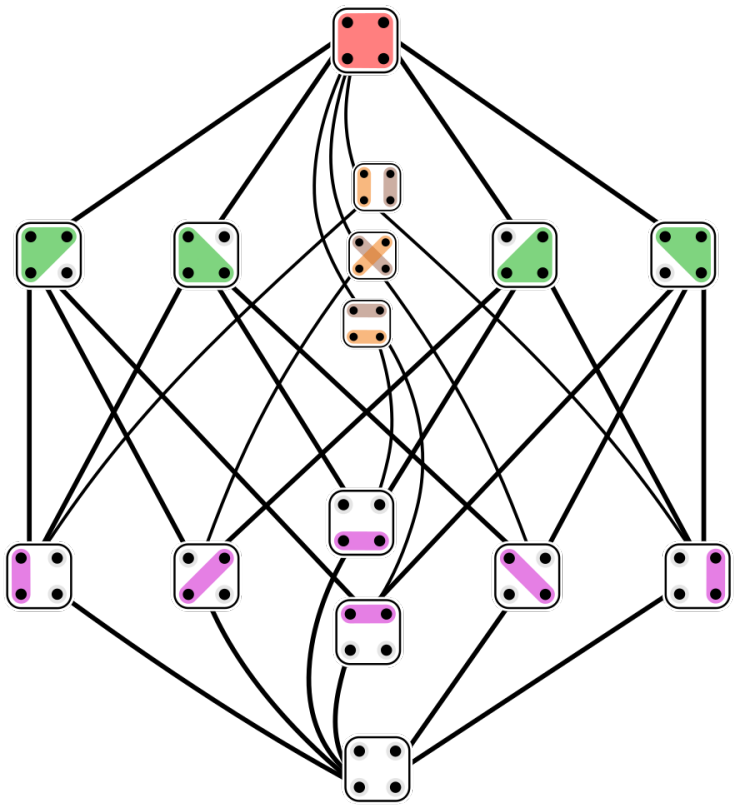


自反 + 反对称 + 传递

B_n: Boolean algebra of degree n

Partitions of a Set; Ordered by **Refinement (细化/精化)**





$$\alpha \leq \beta \iff \forall A \in \alpha\ \exists B \in \beta\ A \subseteq B$$

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对于任意 $x, y, z \in X$:

(1) $x \prec y$, $x = y$, $y \prec x$ 三者中至多有一个成立;

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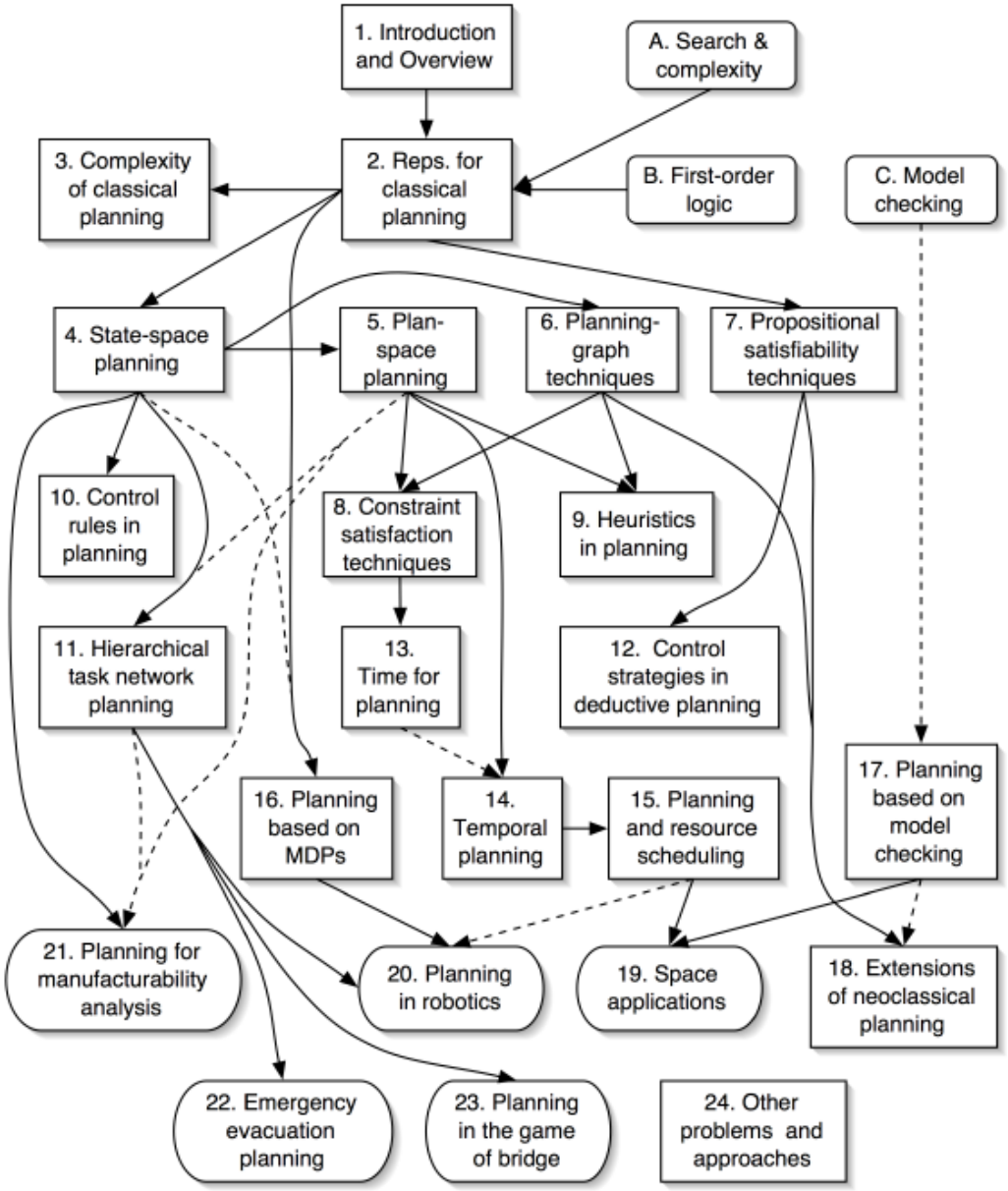
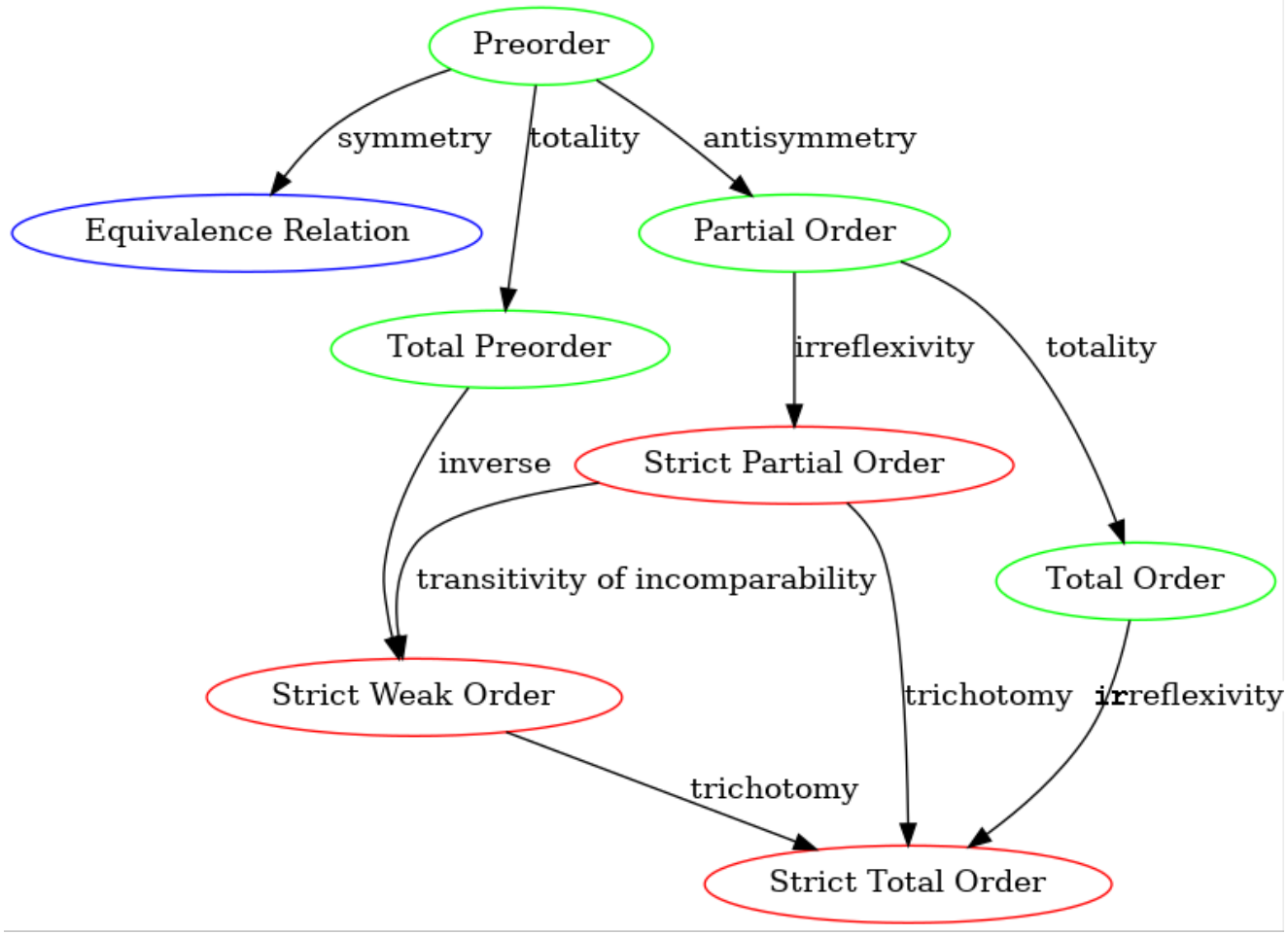

1. Introduction and Overview
A. Search & complexity
3. Complexity of classical planning
2. Reps. for classical planning
B. First-order logic
C. Model checking
4. State-space planning
5. Plan-space planning
6. Planning-graph techniques
7. Propositional satisfiability techniques
10. Control rules in planning
8. Constraint satisfaction techniques
9. Heuristics in planning
11. Hierarchical task network planning
13. Time for planning
12. Control strategies in deductive planning
17. Planning based on model checking
16. Planning based on MDPs
14. Temporal planning
15. Planning and resource scheduling
21. Planning for manufacturability analysis
20. Planning in robotics
19. Space applications
18. Extensions of neoclassical planning
22. Emergency evacuation planning
23. Planning in the game of bridge
24. Other problems and approaches

反自反 + 传递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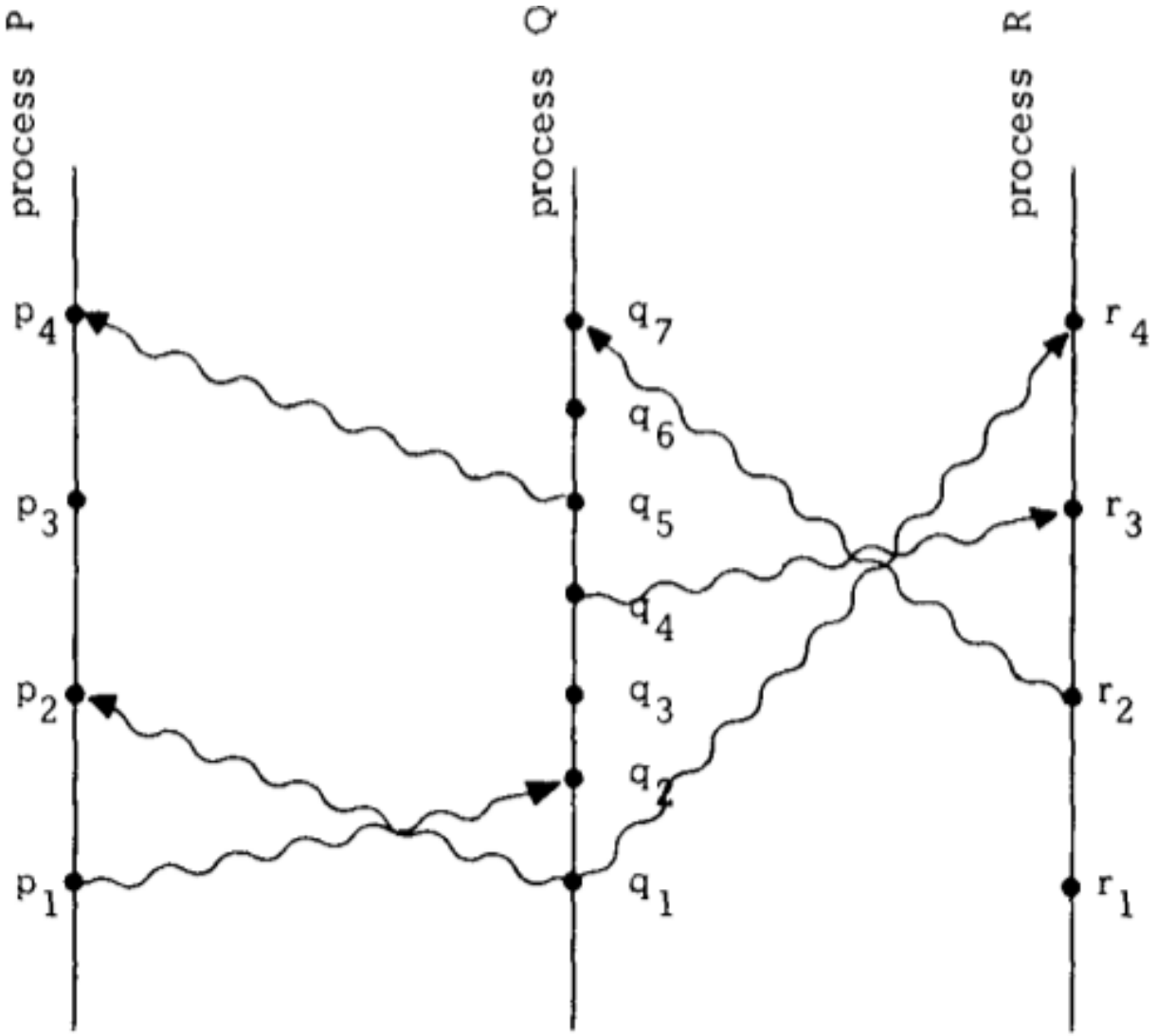
process P
process Q
process R
p4
p3
p2
p1
q7
q6
q5
q4
q3
q2
q1
r4
r3
r2
r1

Definition (事件间的 “先于” 关系 (“Happened Before” Relation on Events))

令 E 是表示事件的集合。

$\rightarrow \subseteq E \times E$ 是 E 上的“先于”关系, 如果它是满足以下条件的最小关系:

- 对任何事件 $a \in E$, $\neg(a \rightarrow a)$;
- 如果 a, b 事件属于同一个进程且 a 在 b 之前发生, 则 $a \rightarrow b$;
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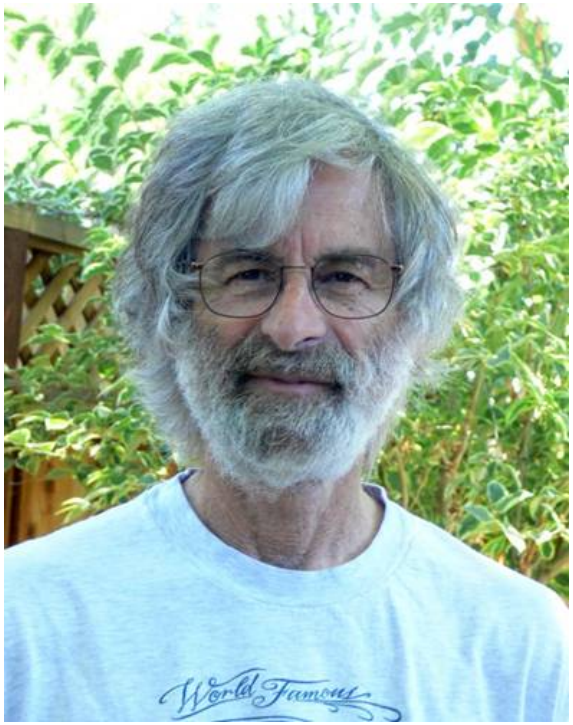

Leslie Lamport (1941 ∼)

For "fundamental contributions to the theory and practice of distributed and concurrent systems, notably the invention of concepts such as causality and logical clocks, safety and liveness, replicated state machines, and sequential consistency"

— *Turing Award (2013)*

Time, Clocks, and the Ordering of Events in a Distributed System

Leslie Lamport
Massachusetts Computer Associates, Inc.

The concept of one event happening before another in a distributed system is examined, and is shown to define a partial ordering of the events. A distributed algorithm is given for synchronizing a system of logical clocks which can be used to totally order the events.

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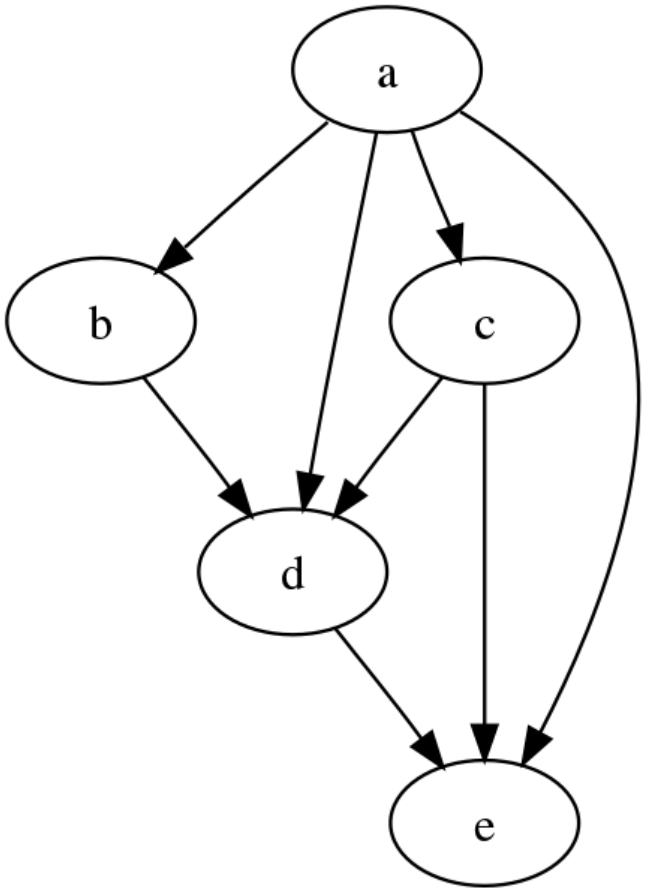

有向无环图 (DAG; Directed Acyclic Graph)

Definition (全序关系 (Total Order))

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全序关系 = 偏序关系 + 连接性 $\quad (4) \Longrightarrow (1)$

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$$(\mathbb{R}, \leq) \qquad (\mathbb{R}, \geq)$$

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(3) 半连接性 (Semi-Connex):

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$$(\mathbb{R}, <) \qquad (\mathbb{R}, >)$$

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$$(\mathbb{R}, <) \qquad (\mathbb{R}, >)$$

严格全序关系 = 严格偏序关系 + 半连接性

Theorem

设 $\prec \subseteq X \times X$ 是 X 上的严格全序关系。

$\prec$ 满足三歧性 *(Trichotomous)*:

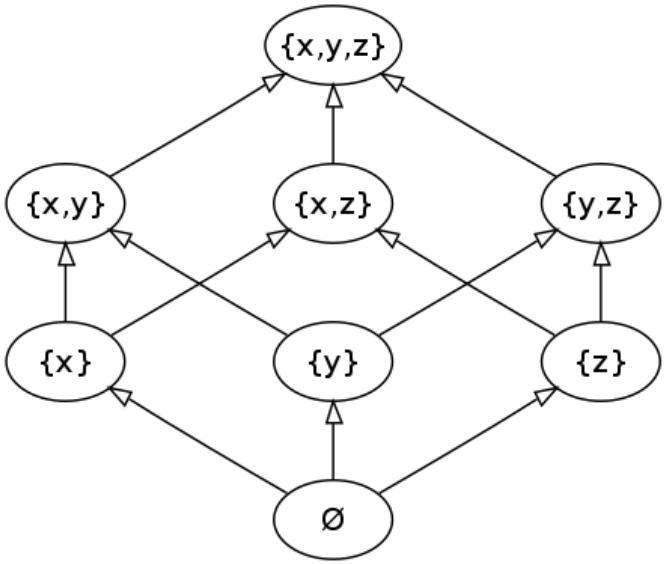
$$\forall x, y \in X \ (\textit{exactly one of}\ x \prec y, x = y,\ \textit{or}\ y \prec x\ \textit{holds})$$

Definition

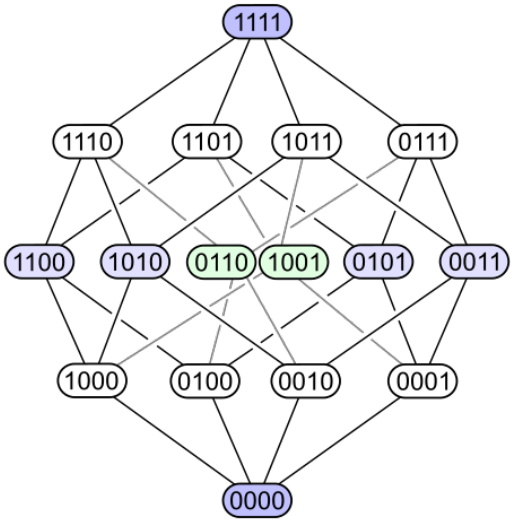
设 $(X, \preceq)$ 是偏序集。对任意 $a, b \in X$, 如果

$$a \prec b \wedge (\forall c \in X\ (c \neq a \wedge c \neq b) \rightarrow \neg(a \preceq c \preceq b))$$

则称 a 被 b 覆盖 (covered)。



哈斯图 (Hasse diagram)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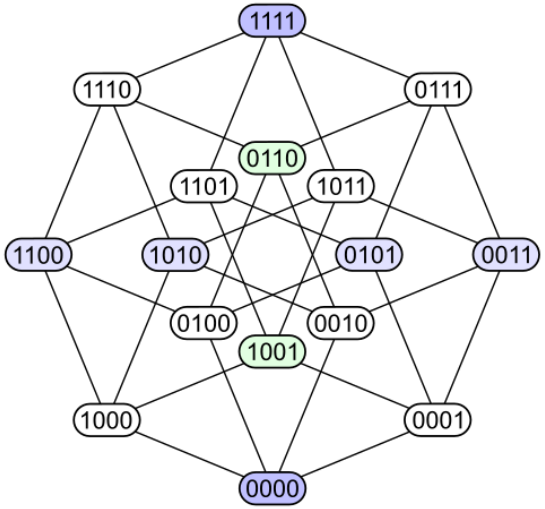
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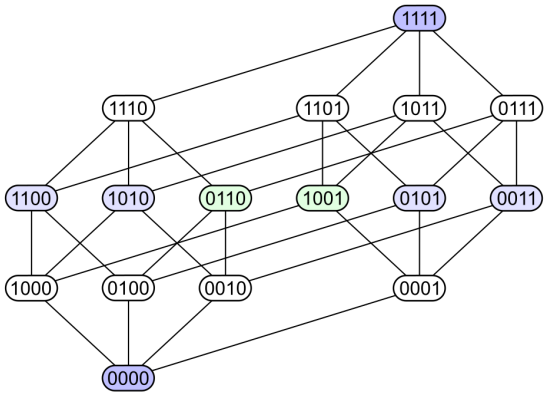
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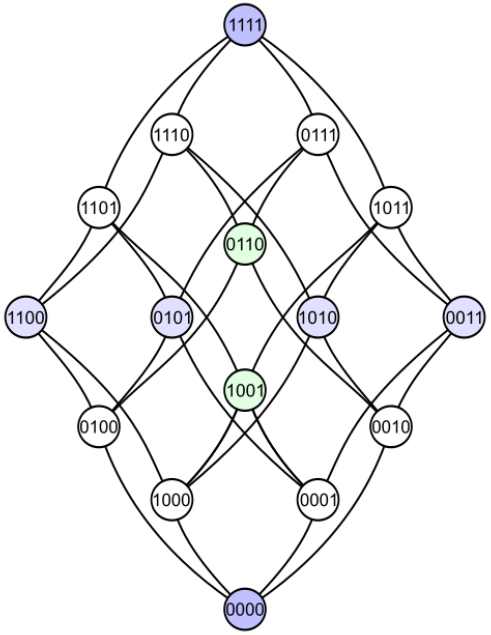
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Definition (极大元与极小元 (Maximal/Minimal))

令 $\preceq \subseteq X \times X$ 是 X 上的偏序。设 $a \in X$,

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Q : 极大/极小元是否一定存在? 如果存在, 是否唯一?

$$(\mathbb{R}, \leq)$$

$$(\mathbb{R}, \leq)$$

无极大元、无极小元

$$(\mathbb{R}, \leq)$$

无极大元、无极小元

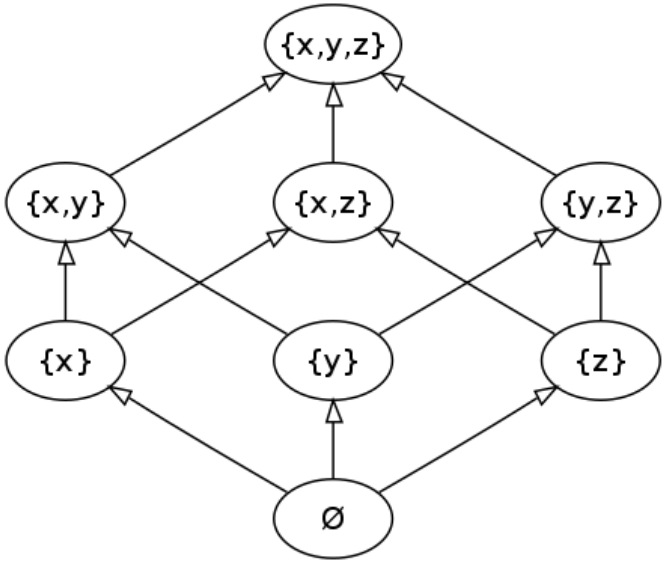
$$(\mathbb{N}, \leq)$$

$$(\mathbb{R}, \leq)$$

无极大元、无极小元

$$(\mathbb{N}, \leq)$$

无极大元、有唯一极小元 0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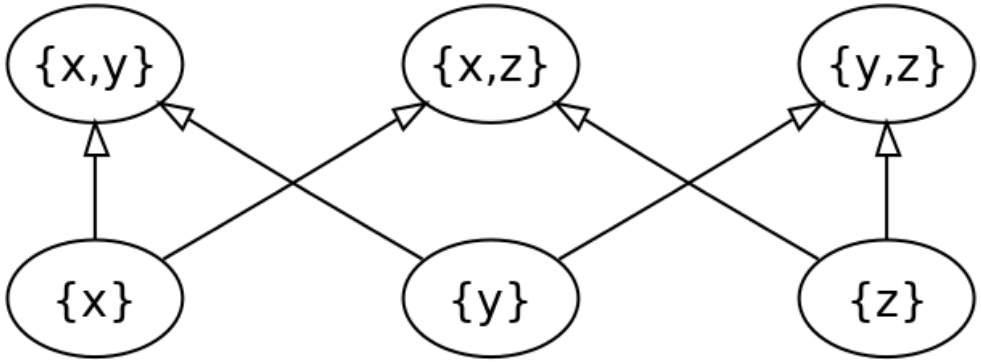
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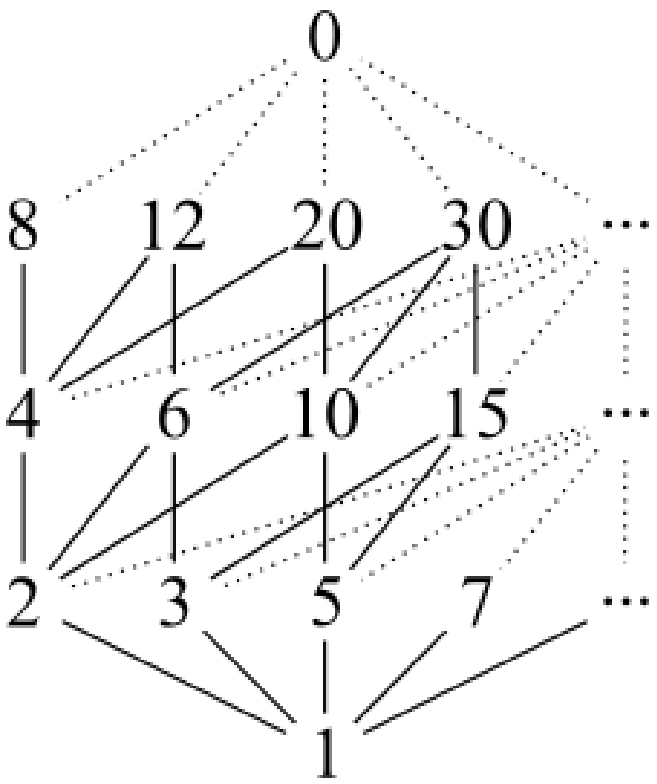

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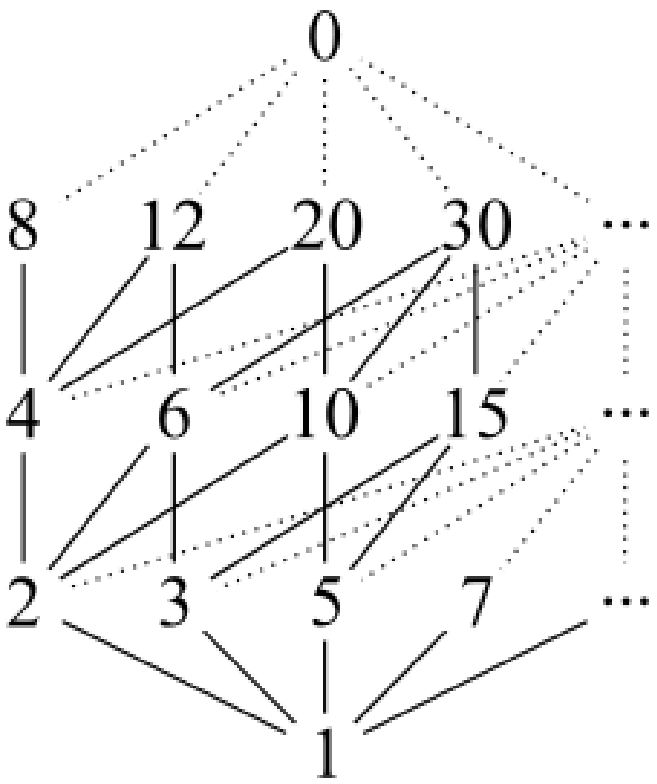

无极大元、有无穷多个极小元 (所有的素数)

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Theorem

偏序集 $(X, \preceq)$ 如果有最大元或最小元，则它们是唯一的。

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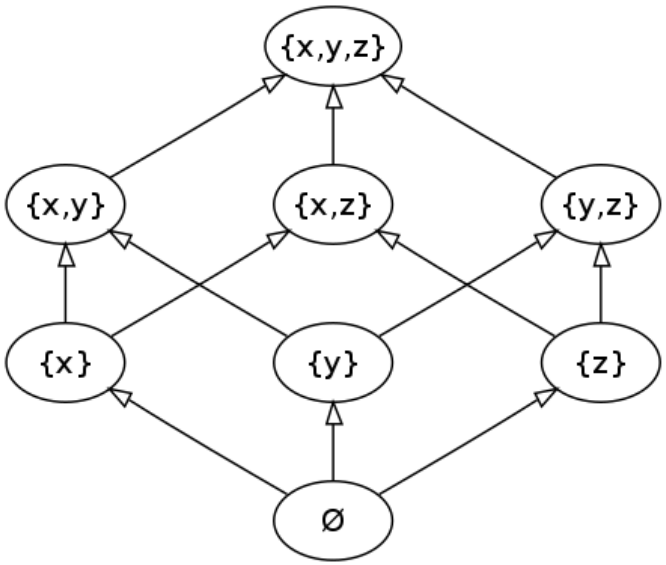
$$x \preceq y \wedge y \preceq x \implies x = y$$

(反对称性)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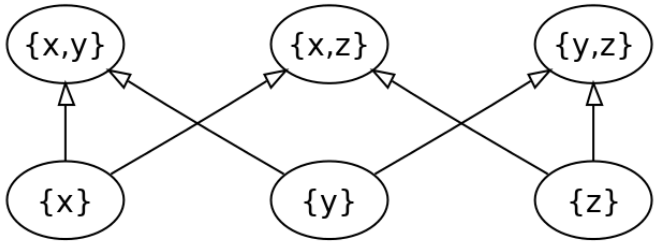
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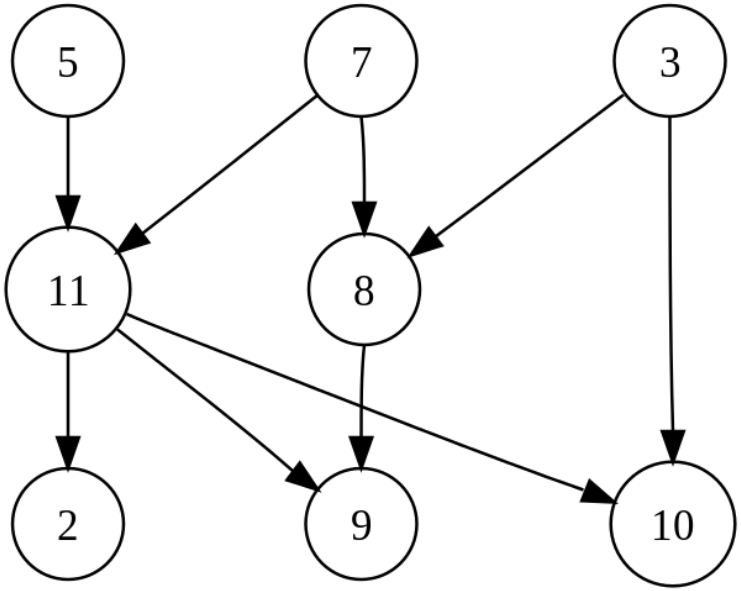
Definition (线性拓展 (Linear Extension))

设 $(X, \preceq)$ 是偏序集, $(X, \preceq')$ 是全序集。如果

$$\forall x, y \in X \; x \preceq y \to x \preceq' y,$$

则称 $\preceq'$ 是 $\preceq$ 的线性拓展。



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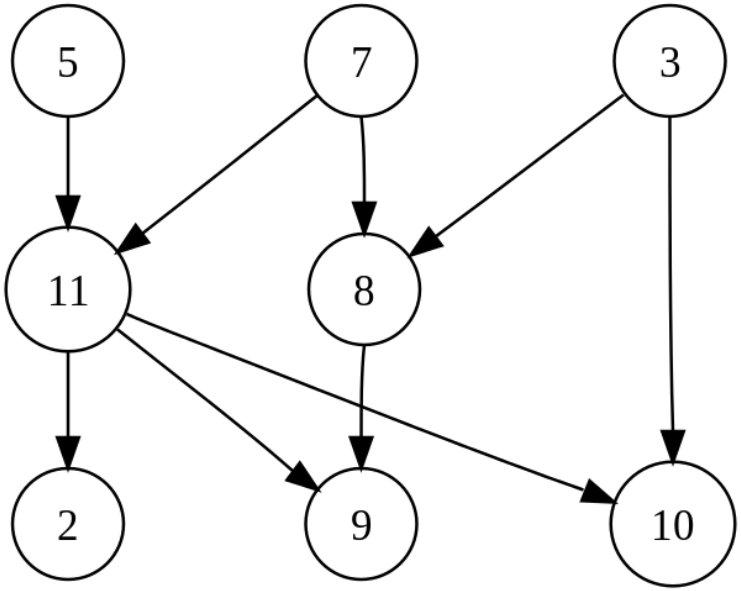
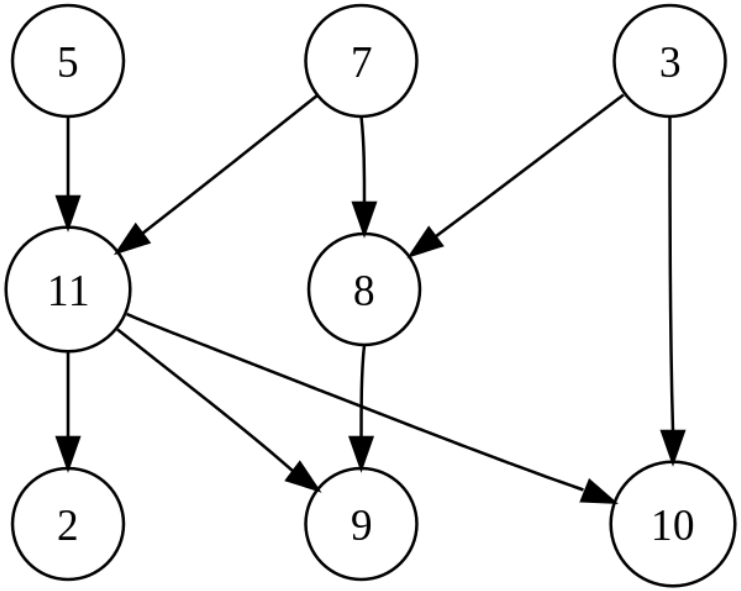


$5, 7, 3, 11, 8, 2, 9, 10$ $\quad$ $3, 5, 7, 8, 11, 2, 9, 10$

Theorem

设 $(X, \preceq)$ 是偏序集且 X 是有限集，则 $\preceq$ 的线性拓展必定存在。



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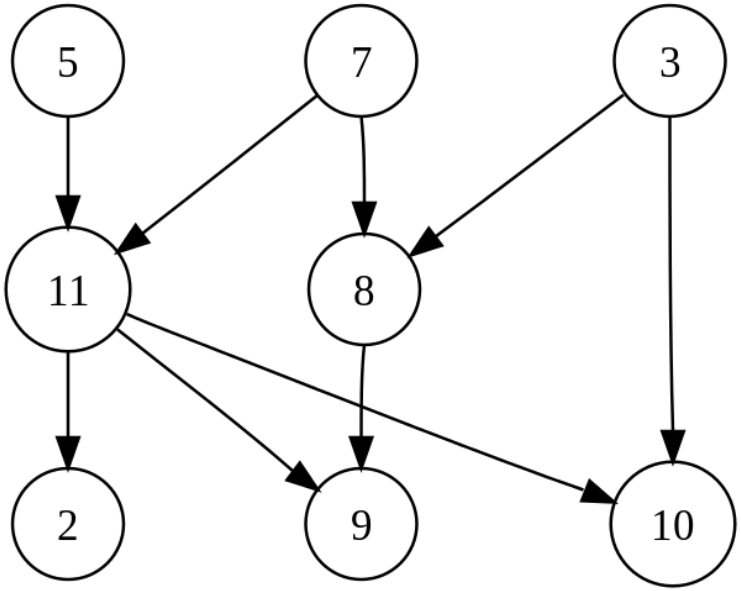

Topological sorting: iteratively remove minimal elements

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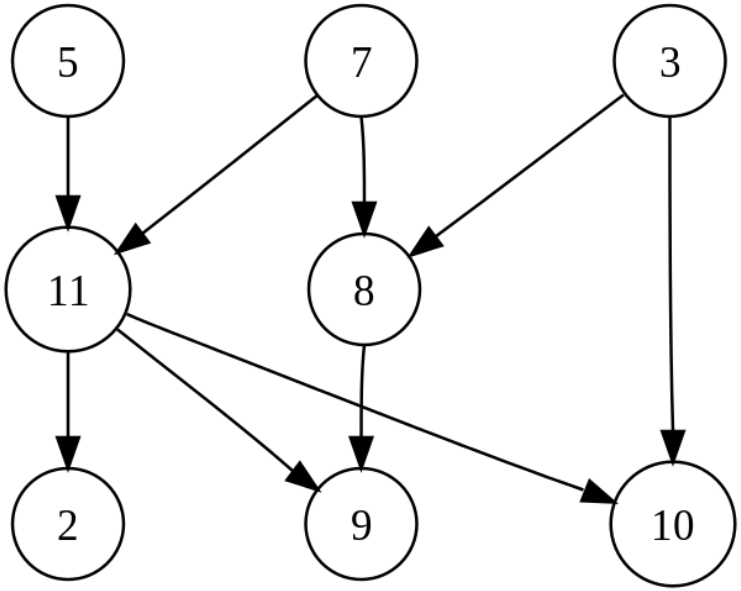

Topological sorting: iteratively remove minimal elements

Theorem

设 $(X, \preceq)$ 是偏序集且 X 是有限集，则极小元 (极大元) 一定存在。

Definition (上界 (Upper Bound) 与下界 (Lower Bound))

设 $(X, \preceq)$ 是偏序集。对于 $Y \subseteq X$, 如果

$$\exists x \in X \ \forall y \in Y \ y \preceq x,$$

则称 x 为 Y 的上界。

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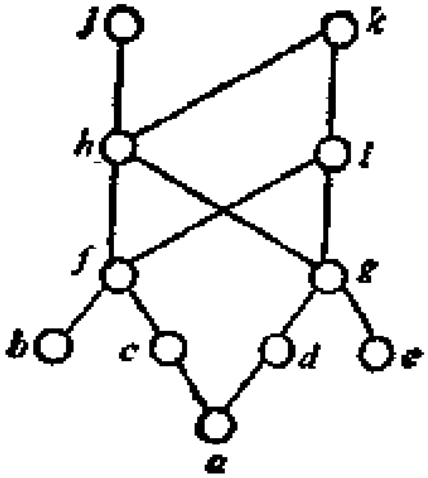

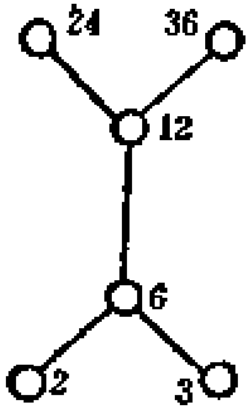

Definition (最小上界 (Least Upper Bound; LUB); 上确界 (Supremum; $\sup(Y)$))

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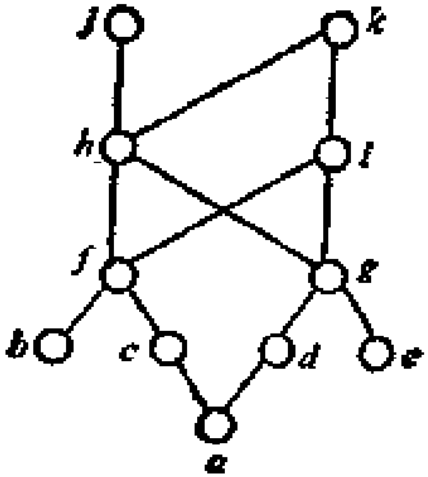

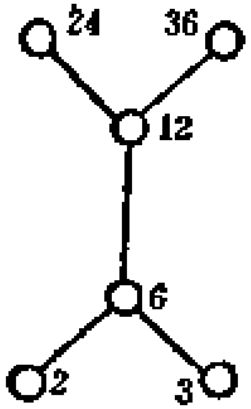

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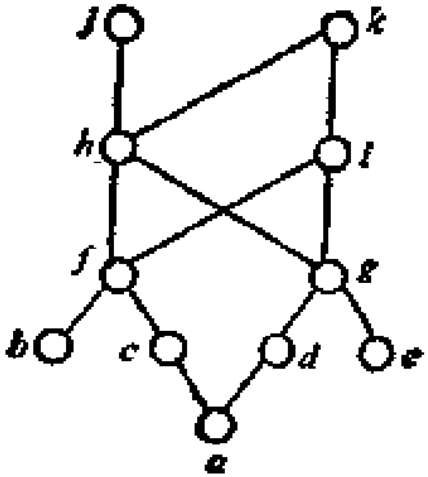

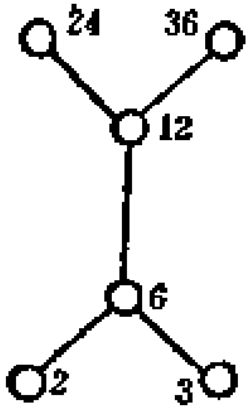

Definition (最大下界 (Greatest Lower Bound; GLB); 下确界 (Infimum; $\inf(Y)$))

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如下 $(A, \leq)$ 有上确界吗?

$$A = \{p \in \mathbb{Q} \mid p > 0 \wedge p^2 < 2\}$$

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在 $(\mathbb{R}, \leq)$ 中, A 有上确界: $\sup(A) = \sqrt{2}$

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在 $(\mathbb{Q}, \leq)$ 中, A 无上确界

Theorem

集合 A 在 $(\mathbb{Q}, \leq)$ 中无上确界。

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$$A = \{p \in \mathbb{Q} \mid p > 0 \wedge p^2 < 2\}$$

Proof.

$$B = \{p \in \mathbb{Q} \mid p > 0 \wedge p^2 > 2\}$$

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只需证明: A 无最大元、B 无最小元

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$$\forall p \in A\ \exists q \in A\ p < q \qquad \forall p \in B\ \exists q \in B\ p > q$$

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$$p \in A \implies q \in A \wedge p < q$$

Theorem

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Definition (实数 ($\mathbb{R}$))

如果集合 $A \subseteq \mathbb{Q}$ 满足:

(1) $A \neq \emptyset$ 且 $A \neq \mathbb{Q}$;

(2) A 是向下封闭的:

$$\forall p \in A, q \in \mathbb{Q}\ (q < p \to q \in A)$$

(3) A 没有最大元:

$$\forall p \in A\ \exists q \in A\ p < q$$

则称 A 是**戴德金分割** (Dedekind cut), A 所对应的**实数**为 $\sup(A)$。

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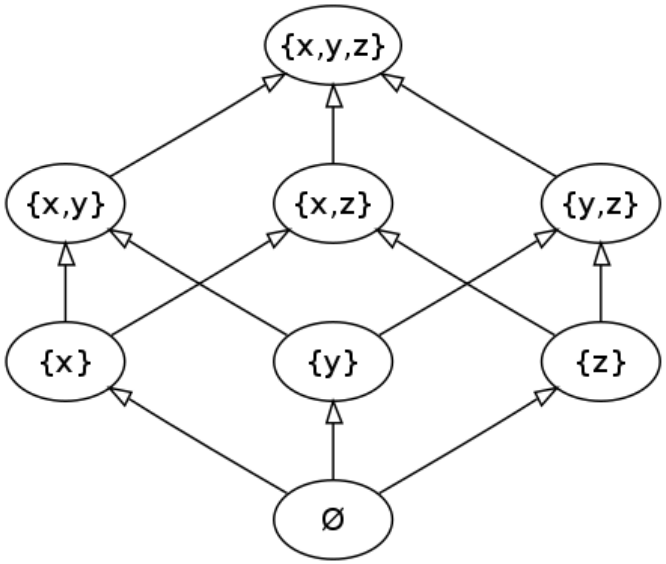

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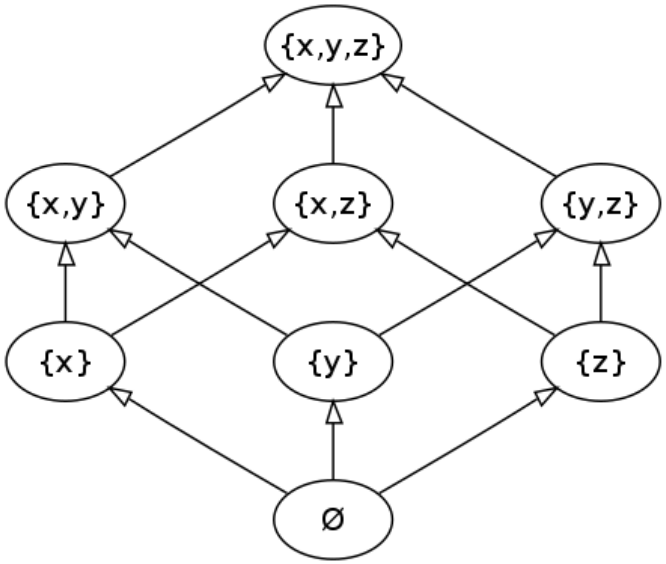

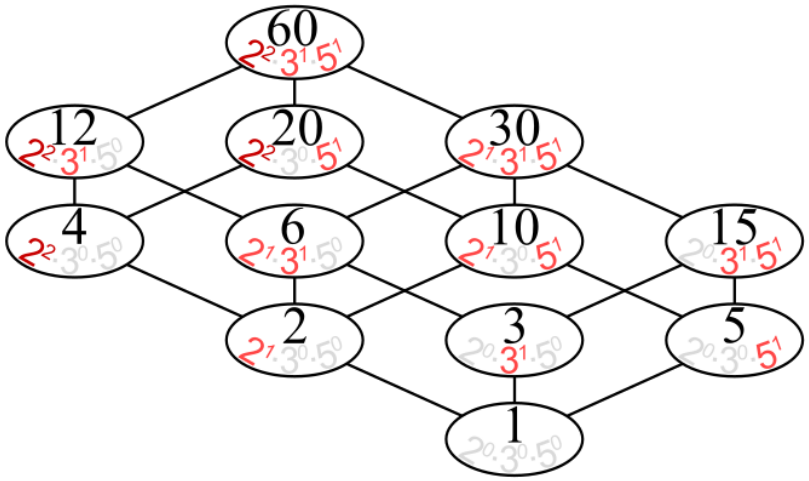

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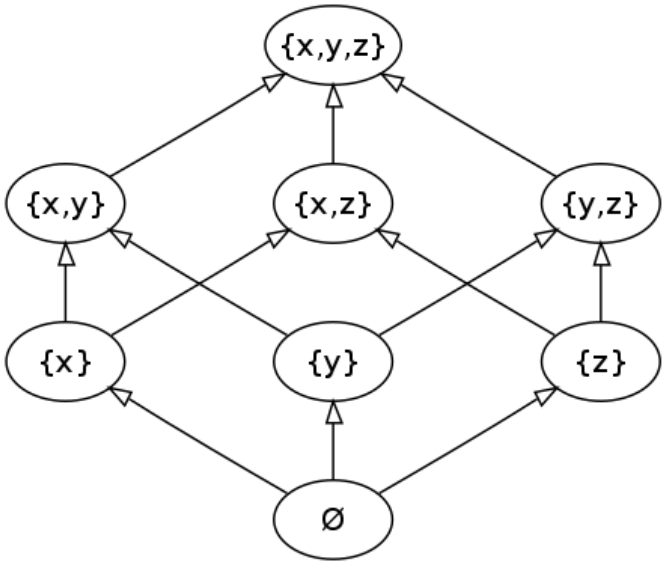

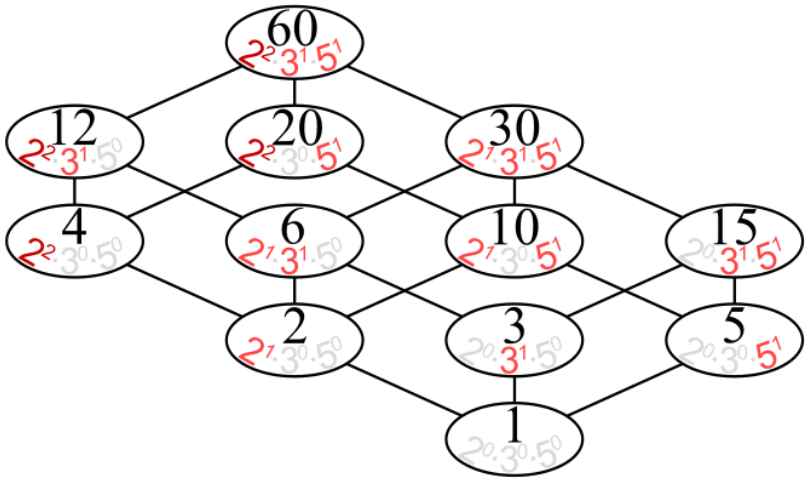

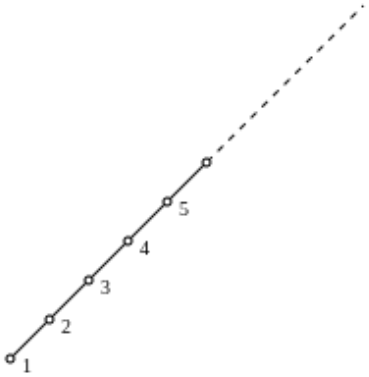

"LET'S STOP HERE, SO WE HAVE
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Definition (良序 (Well-Ordering))

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数学归纳法 良序集

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Dilworth

Equivalent theorems

Definition (可比与不可比 (Comparable/Incomparable))

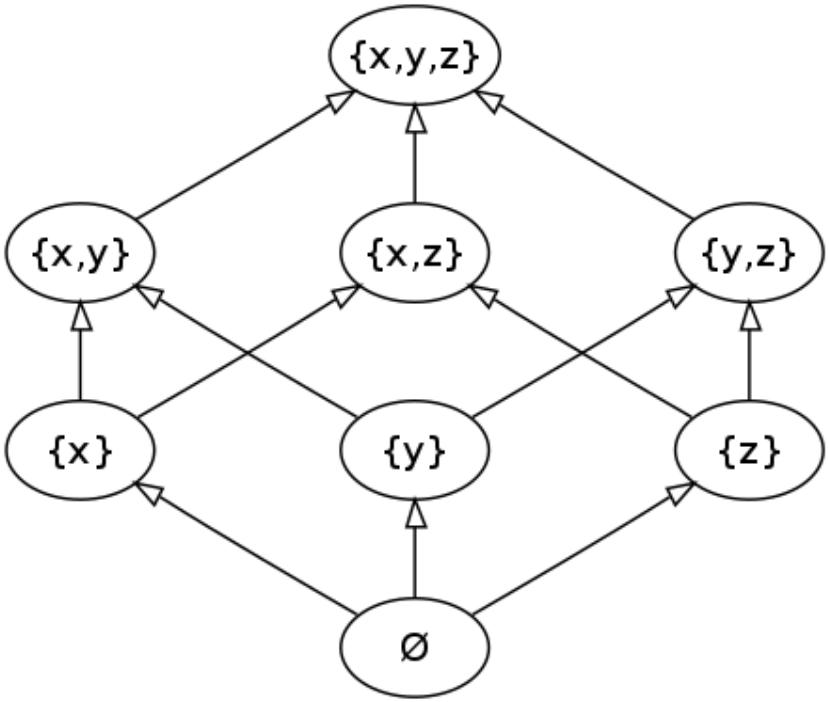
设 $(X, \preceq)$ 是偏序集。对 $a, b \in X$,

可比的 (Comparable):

$$a \preceq b \vee b \preceq a$$

不可比的 (Incomparable):

$$\neg(a \preceq b \vee b \preceq a)$$

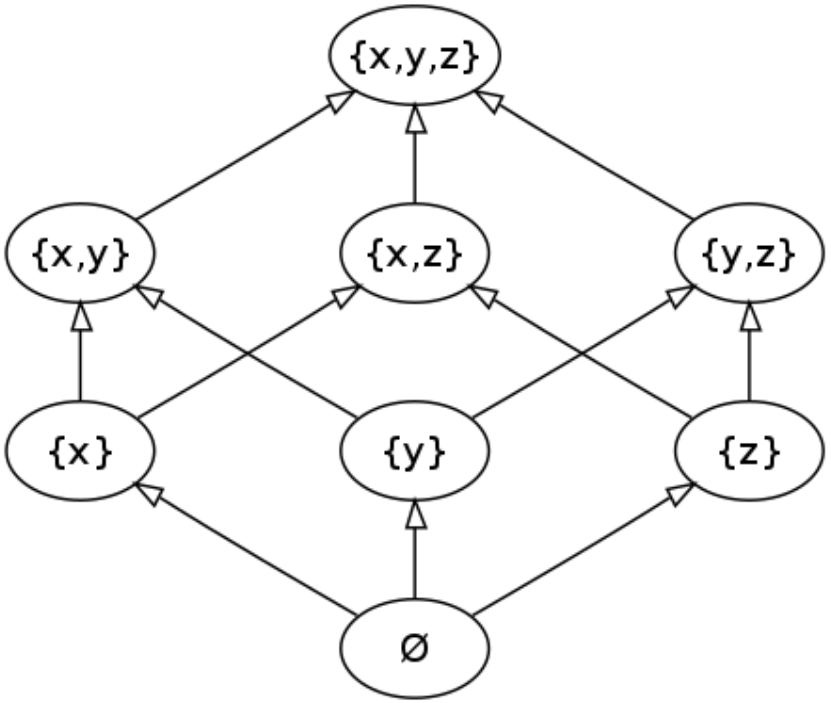


Definition (链与反链 (Chain; Antichain))

设 $(X, \preceq)$ 是偏序集。

- 设 $S \subseteq X$ 且 S 中元素两两可比, 则称 S 是链。
- 设 $S \subseteq X$ 且 S 中元素两两不可比, 则称 S 是反链。

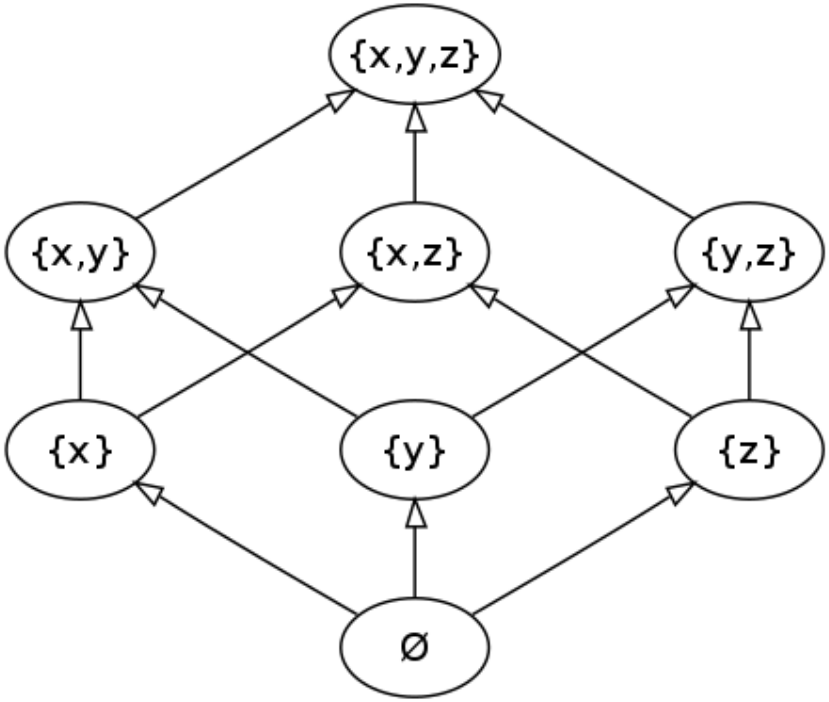
注意: 单元素集合既是链, 也是反链



Definition (宽度与高度 (Width; Height))

设 $(X, \preceq)$ 是偏序集。

- 宽度 (Width): X 中最大反链的大小
- 高度 (Height): X 中最大链的大小

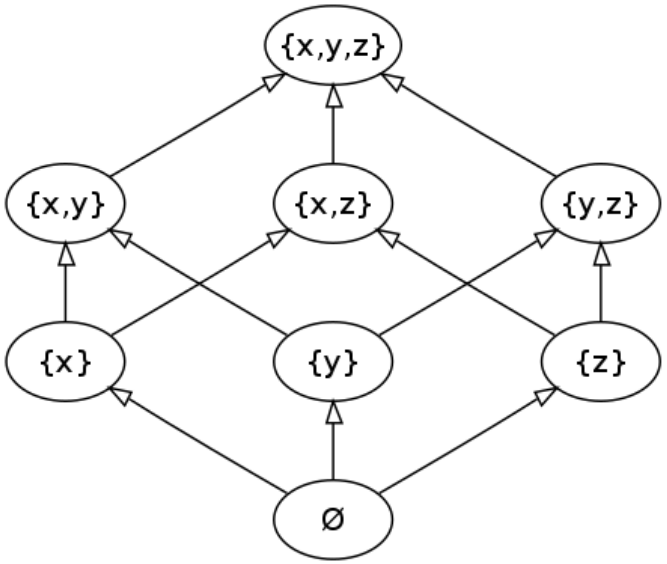


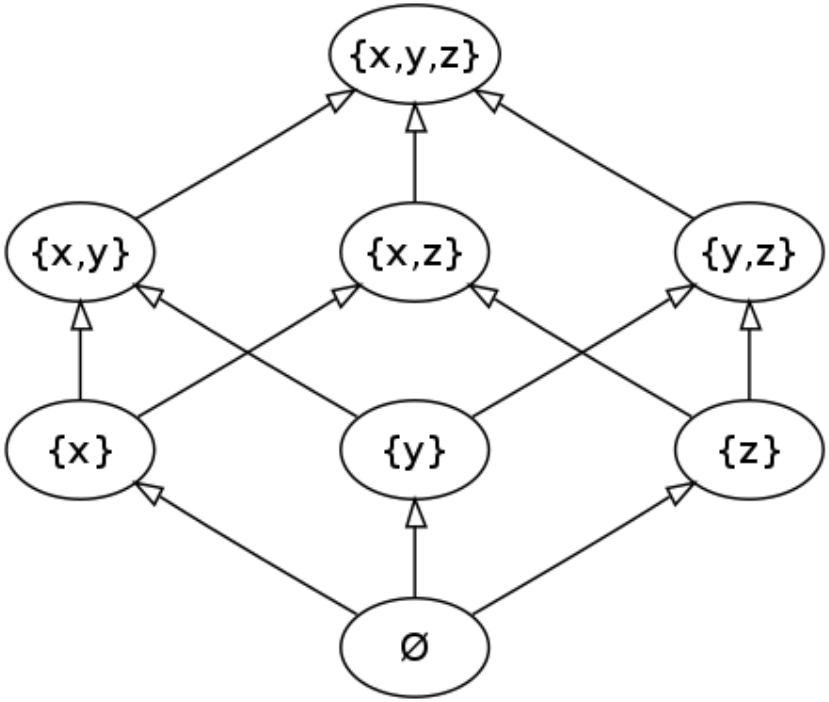
Definition (链分解 (Chain Decomposition))

设 $(X, \preceq)$ 是偏序集。对于 $C_1, C_2, \ldots, C_k \subseteq X$, 如果

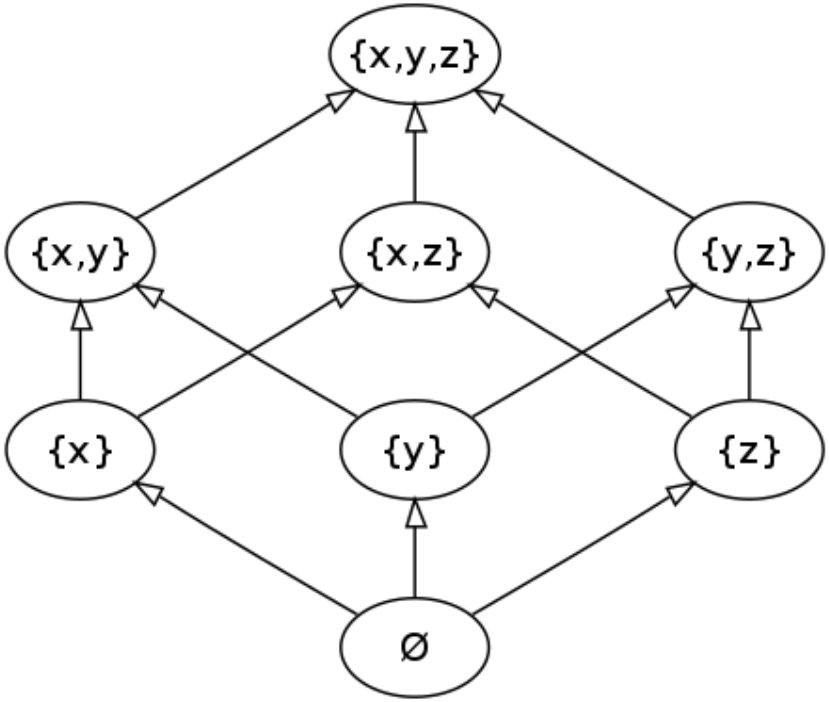
1. 每个 C_i 都是**链**;
2. $C_1, C_2, \ldots, C_k$ **互不相交**;
3. $X = \bigcup_{i=1}^{k} C_i$,

则称 $\{C_1, C_2, \ldots, C_k\}$ 是 X 的**链分解**。




{x,y,z}
{x,y}
{x,z}
{y,z}
{x}
{y}
{z}
∅

$$\{\{x\},\{y\},\{z\}\}$$



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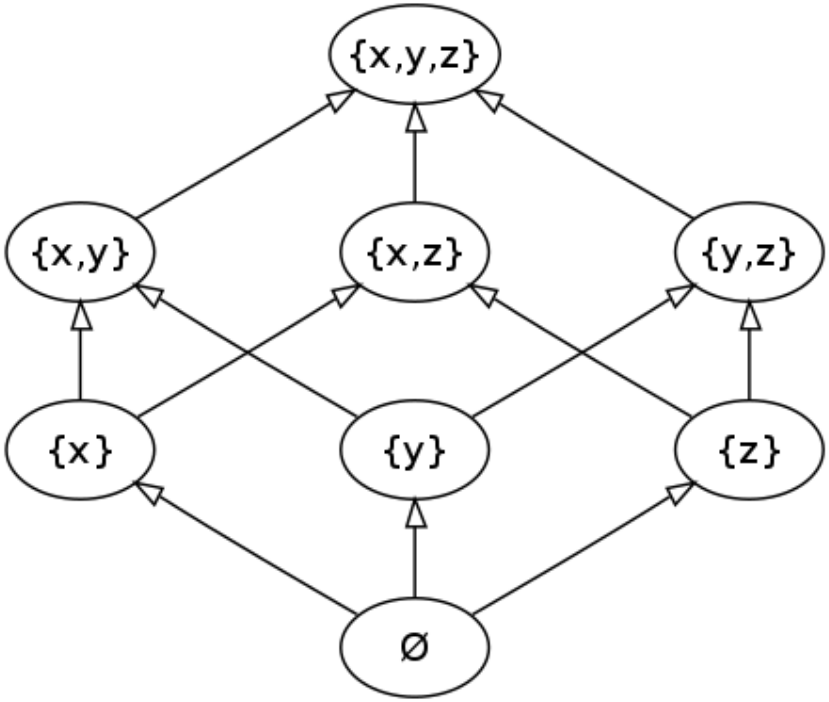

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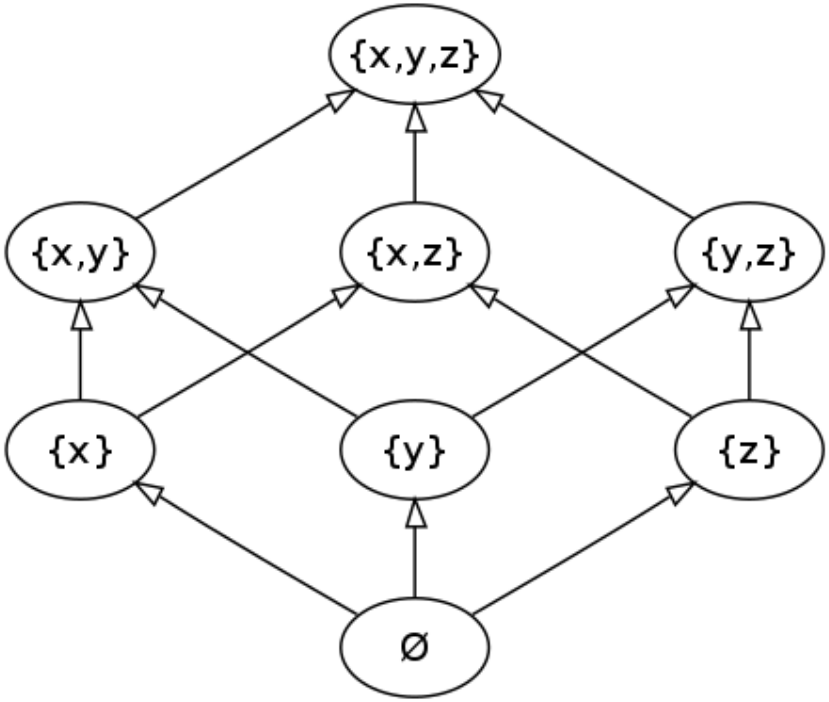
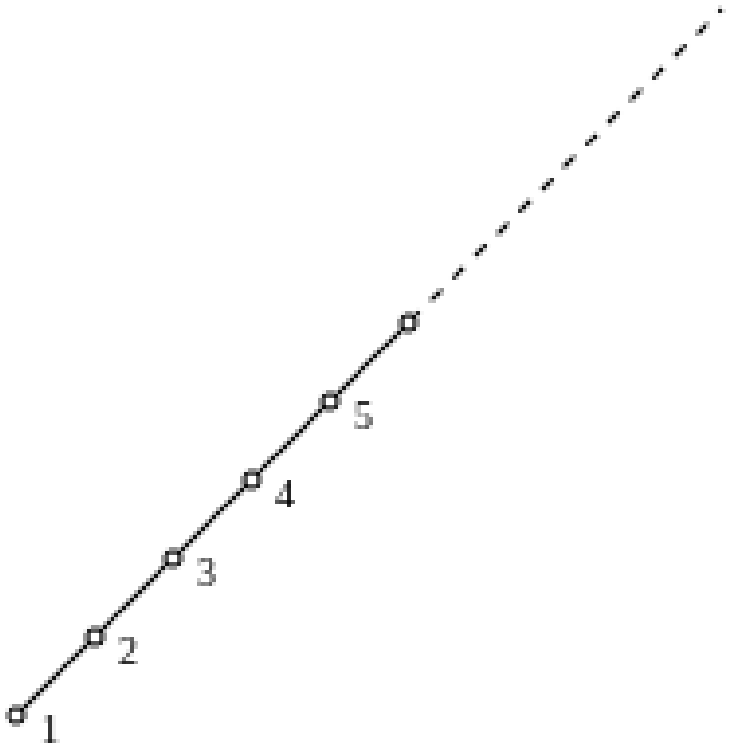
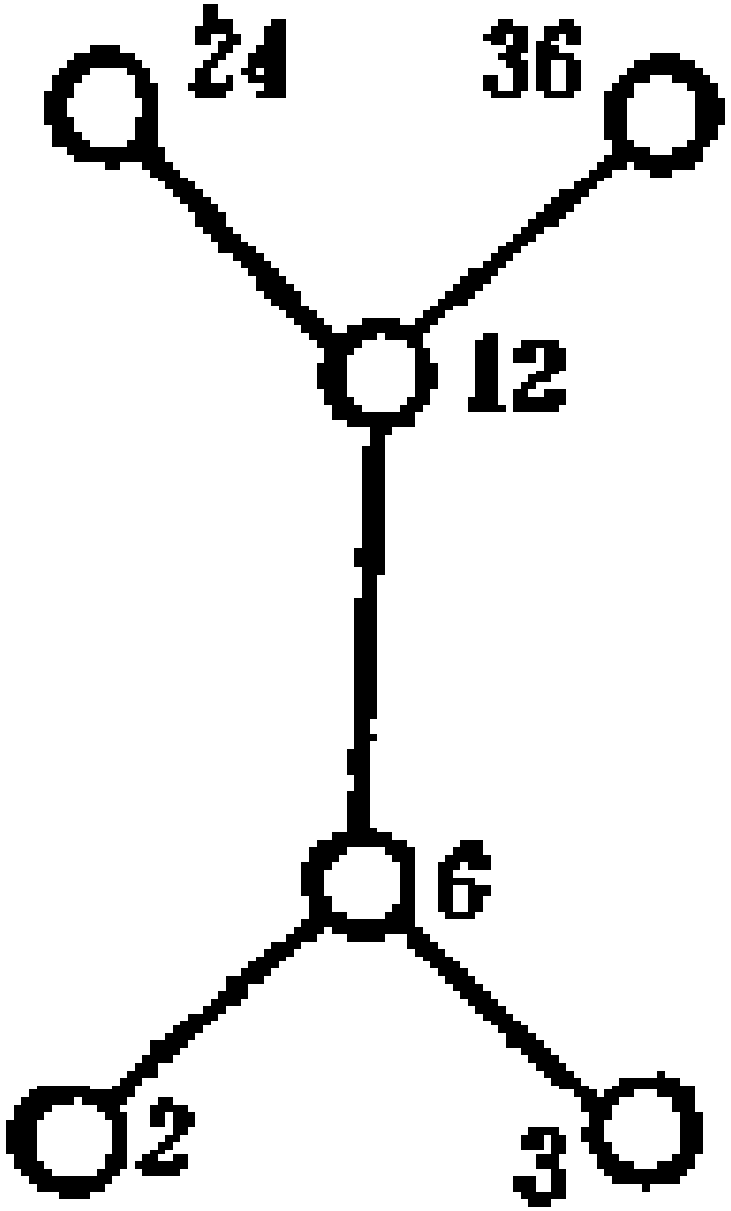
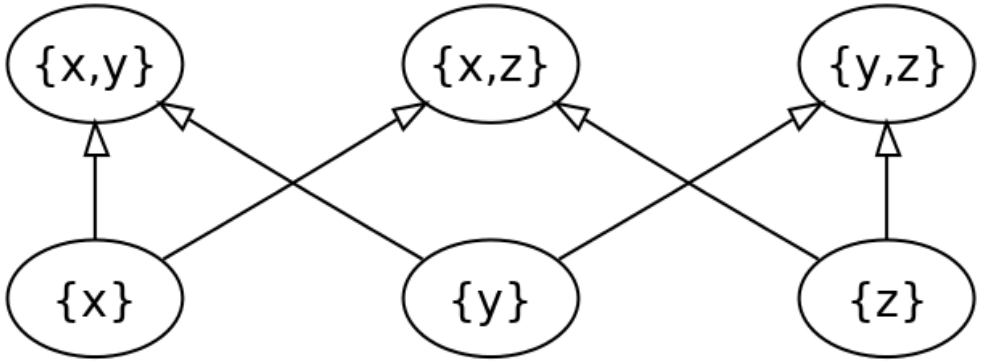


$$\{\{\emptyset,\{x\},\{x,y\},\{x,y,z\}\},\{\{y\},\{y,z\}\},\{\{z\},\{x,z\}\}\}$$


1
2
3
4
5


24
36
12
6
2
3


{x,y}
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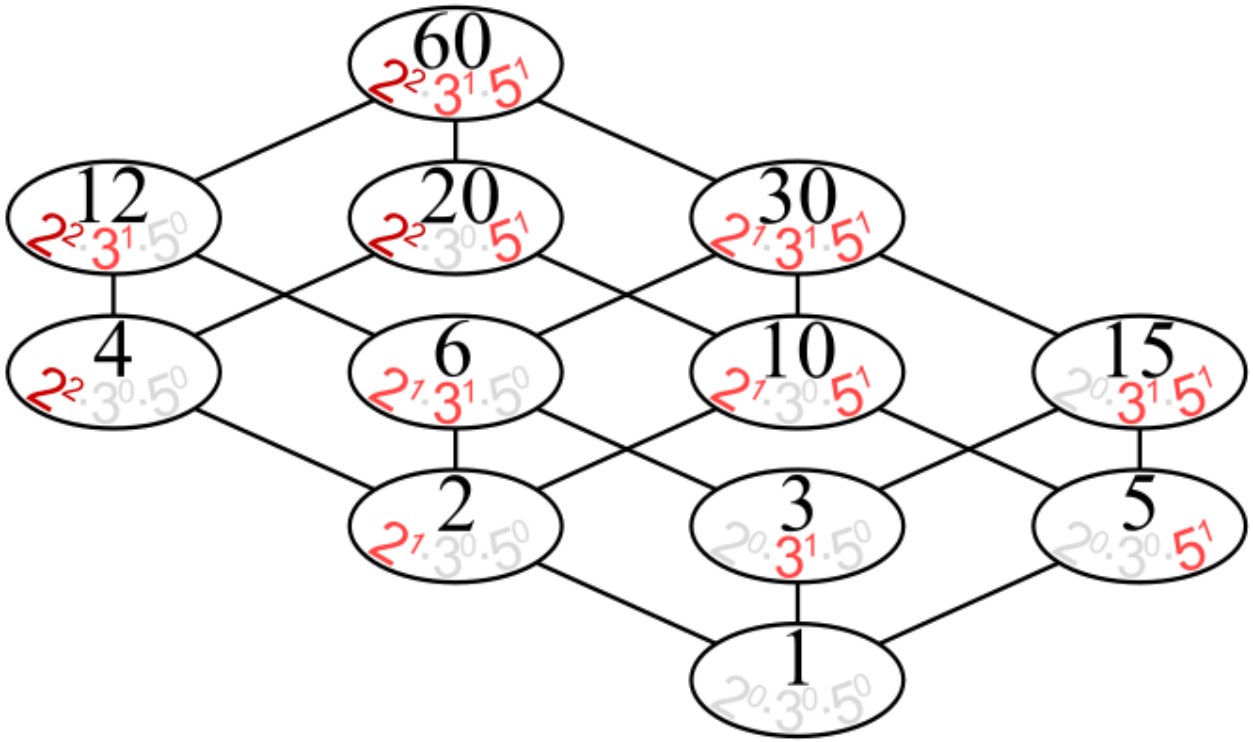
60
$2^2.3^1.5^1$
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$2^2.3^1.5^0$
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$2^1.3^1.5^0$
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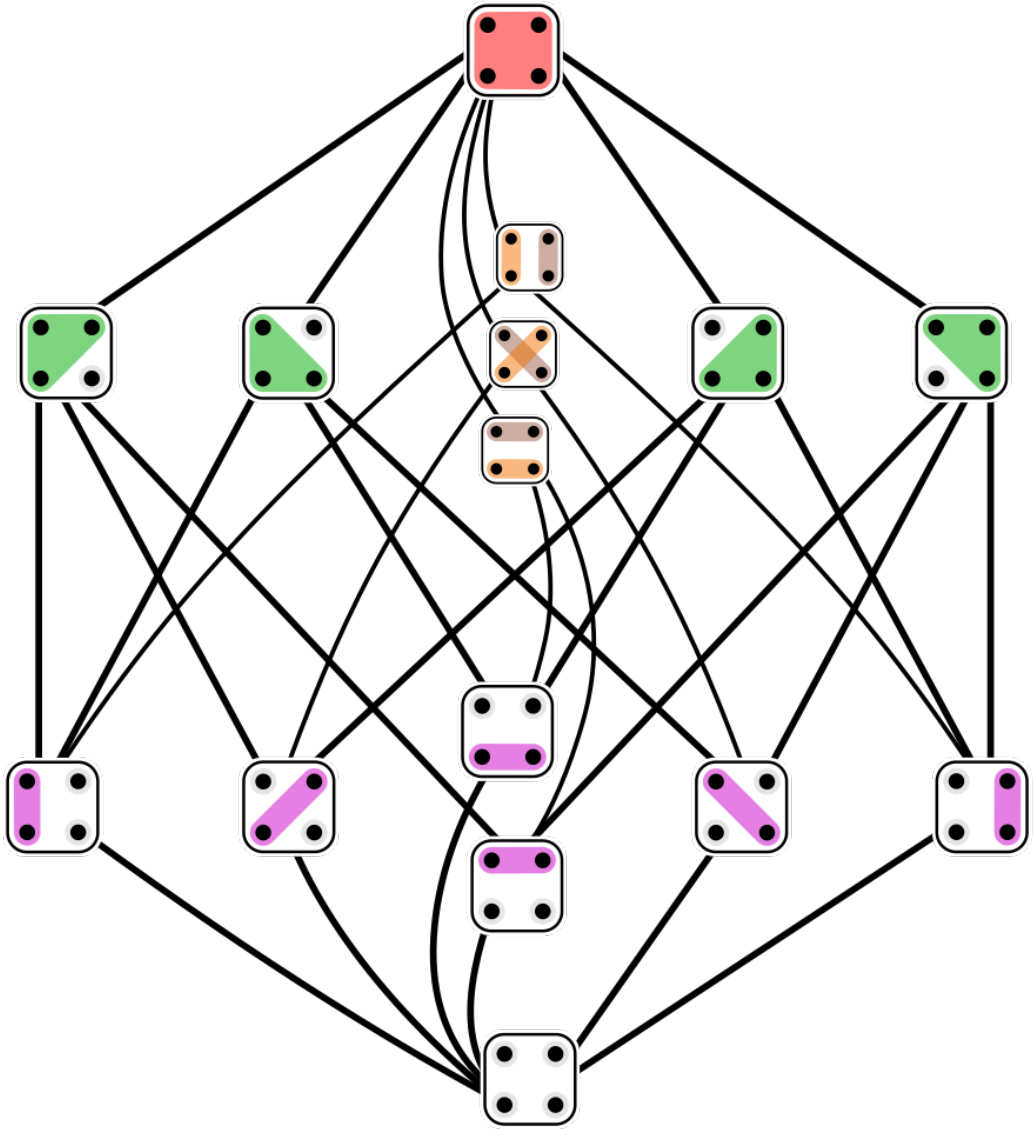

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Antichains: concurrency or independence Chain decompositions: serial pipelines

这将是迄今为止我们在本课程中证明的最难的定理。

精密仪器、环环相扣

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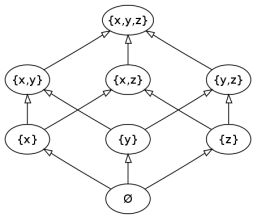

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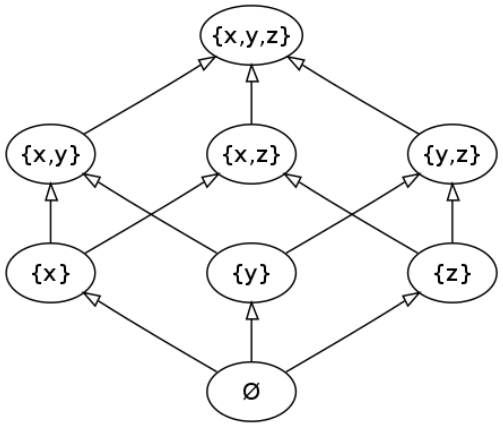

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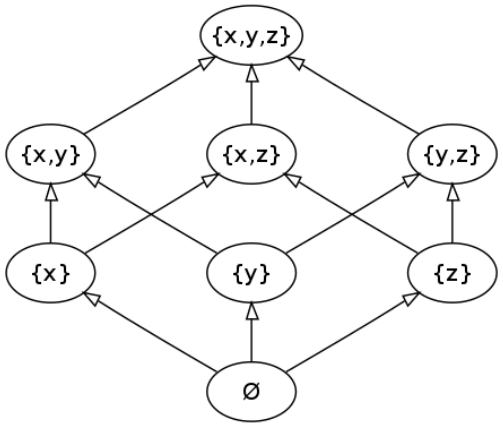

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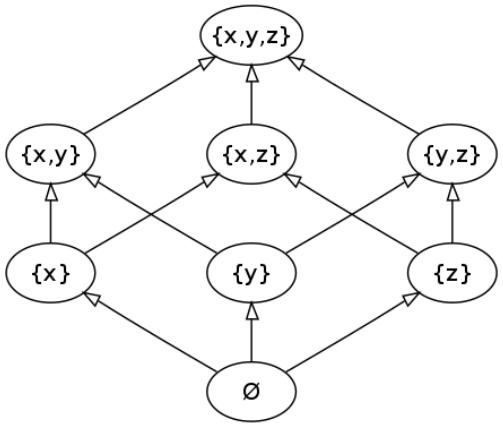

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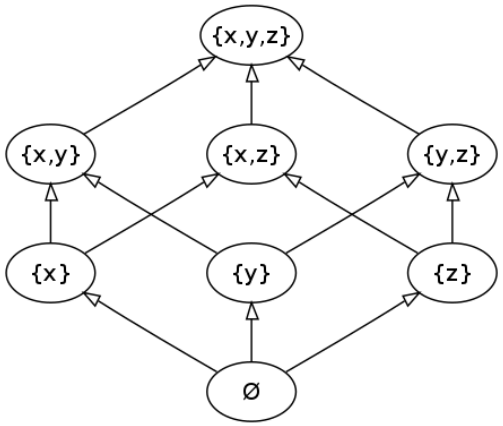

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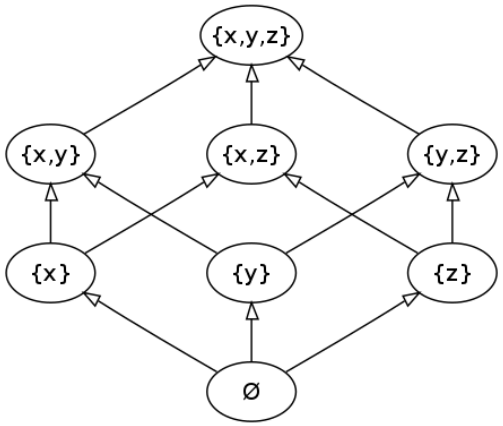

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Each chain contains **exactly one** element of the maximum antichain.

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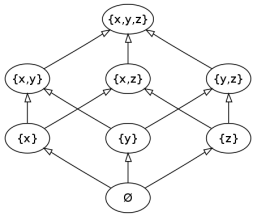
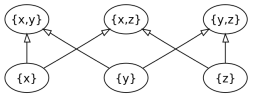
two cases





Lemma ($\leq$: Upper Bound of Chain Decomposition Size)

$$\textit{minimum chain-decomposition size} \leq \mathrm{width}(X).$$

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Inductive Hypothesis: Suppose the theorem holds for all posets of size $2 \leq |X| < n$.

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- Suppose A is a maximum antichain of X with k elements.

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- Suppose A is a maximum antichain of X with k elements.
- We construct a chain decomposition of X with k chains.

□

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Case 1: Every largest antichain is exactly the set of all minimal elements, or exactly the set of all maximal elements.

Case 2: There exists a largest antichain that is neither the set of all minimal elements nor the set of all maximal elements.

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$$C^m : m = x_1 \prec x_2 \prec \cdots \prec x_t$$

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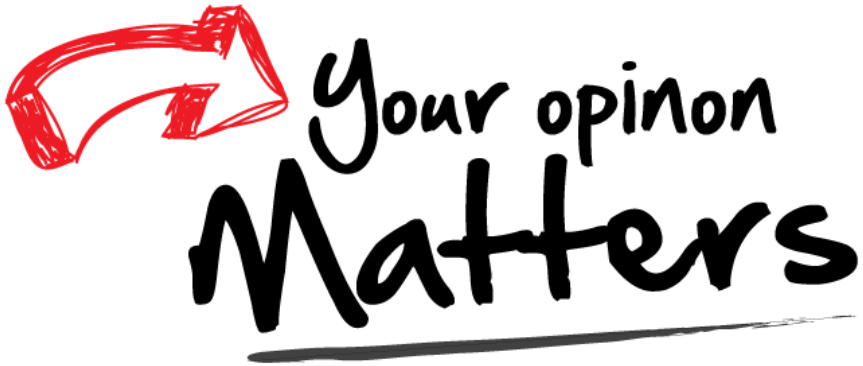
Thank You!



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